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SCHOOL OF ELECTRONICS ENGINEERING

Fall Semester 2024-2025

BECE406E – FPGA Based System Design Lab

REG NO : 23BEC0206

NAME : SRIRAM KP

SLOT : L27 + L28

TASK 4 – Static Timing Analysis

Aim:

i) To design and implement a digital clock that displays time in the format of Hours, Minutes, and Seconds (HH:MM:SS) using the clock divider concept. The output is shown on a seven segment display, and the functionality is verified through simulation and FPGA implementation with relevant screenshots.

ii) To design and implement a synchronous FIFO using Verilog in Quartus and verify its operation through simulation. Perform static timing and power analysis, check for setup and hold time violations, and determine the critical path of the design. Include the RTL view along with screenshots of code and outputs.

Software Details:

Functional Simulation: **Modelsim**

Synthesis & Implementation: **Intel Quartus Prime 23.1**

Hardware Details:

Family: Cyclone IV E

Package: FPGA

Pin count: 780

Core speed grade – 7

Device -EP4CE115F29C7

PART 1)

i) Display the time in the format Hr: Min: Sec using clock divider concept and display the output using seven segment display. Attach screenshots of code and FPGA board implementation.

Verilog RTL Design of Digital Clock with Clock Divider and Seven Segment Display :

```
module real_clock( input clk,input rst,output [6:0] HEX0, output [6:0] HEX1,
   output [6:0] HEX2, output [6:0] HEX3, output [6:0] HEX4,  output [6:0] HEX5,
   output [5:0] sec_out,output [5:0] min_out, output [4:0] hour_out);
reg [25:0] count;
reg tick_1hz;
always @(posedge clk or negedge rst) begin
   if(!rst) begin
      count <= 0;
      tick_1hz <= 0;
   end
   else if(count == 49_999_999) begin
count <= 0;
      tick_1hz <= 1;
   end
   else begin
      count <= count + 1;
      tick_1hz <= 0;
   end
end
reg [5:0] sec;
```

```
reg [5:0] min;
reg [4:0] hour;
always @(posedge clk or negedge rst) begin
   if(!rst) begin
      sec <= 0;
      min <= 0;
      hour <= 0;
   end
   else if(tick_1hz) begin
      if(sec == 59) begin
         sec <= 0;
         if(min == 59) begin
            min <= 0;
            if(hour == 23)
               hour <= 0;
            else
               hour <= hour + 1;
         end
         else
            min <= min + 1;
end
      else
         sec <= sec + 1;
   end
end
assign sec_out  = sec;
assign min_out  = min;
assign hour_out = hour;
```

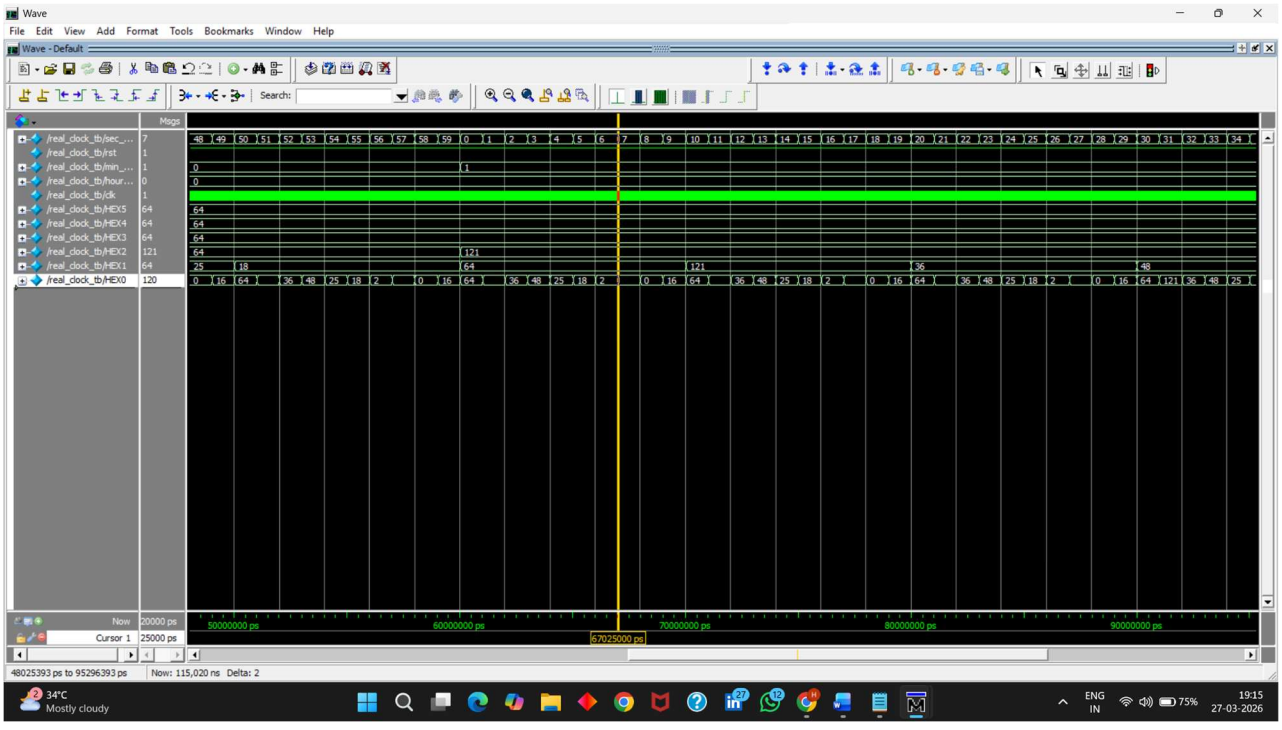
```
wire [3:0] sec_l  = sec % 10;
wire [3:0] sec_h  = sec / 10;
wire [3:0] min_l  = min % 10;
wire [3:0] min_h  = min / 10;
wire [3:0] hour_l = hour % 10;
wire [3:0] hour_h = hour / 10;
hex_to_7seg d0(sec_l, HEX0);
hex_to_7seg d1(sec_h, HEX1);
hex_to_7seg d2(min_l, HEX2);
hex_to_7seg d3(min_h, HEX3);
hex_to_7seg d4(hour_l, HEX4);
hex_to_7seg d5(hour_h, HEX5);
endmodule

module hex_to_7seg(
input [3:0] hex,
output reg [6:0] seg // active low
);
always @(*) begin
   case(hex)
      4'h0: seg = 7'b1000000;
      4'h1: seg = 7'b1111001;
      4'h2: seg = 7'b0100100;
      4'h3: seg = 7'b0110000;
      4'h4: seg = 7'b0011001;
      4'h5: seg = 7'b0010010;
      4'h6: seg = 7'b0000010;
      4'h7: seg = 7'b1111000;
```

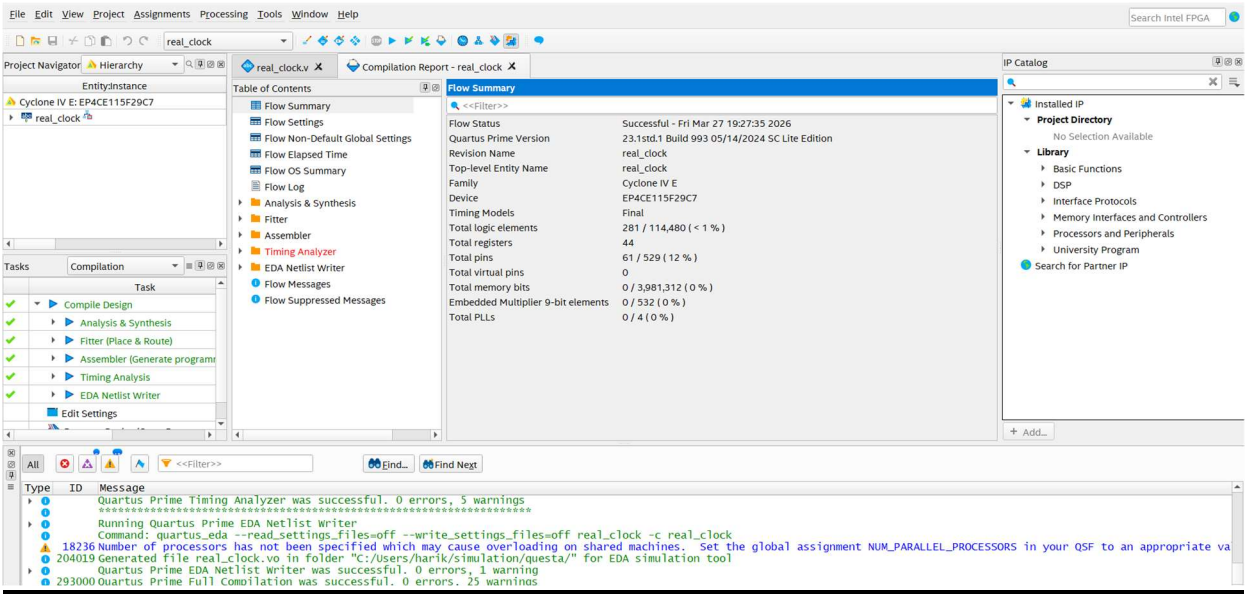
```
        4'h8: seg = 7'b0000000;
        4'h9: seg = 7'b0010000;
        default: seg = 7'b1111111;
    endcase
end
endmodule
`timescale 1ns/1ps
module real_clock_tb;
reg clk;
reg rst;
wire [6:0] HEX0, HEX1, HEX2, HEX3, HEX4, HEX5;
wire [5:0] sec_out, min_out;
wire [4:0] hour_out;
real_clock uut (
    .clk(clk),
    .rst(rst),
    .HEX0(HEX0),
    .HEX1(HEX1),
    .HEX2(HEX2),
    .HEX3(HEX3),
    .HEX4(HEX4),
 .HEX5(HEX5),
    .sec_out(sec_out),
    .min_out(min_out),
    .hour_out(hour_out));
always #5 clk = ~clk;
initial begin
    clk = 0;
```

```
    rst = 0;
    #20 rst = 1;
    #5000;
    $stop;
end
endmodule
```

Functional Simulation of Digital Clock using ModelSim :

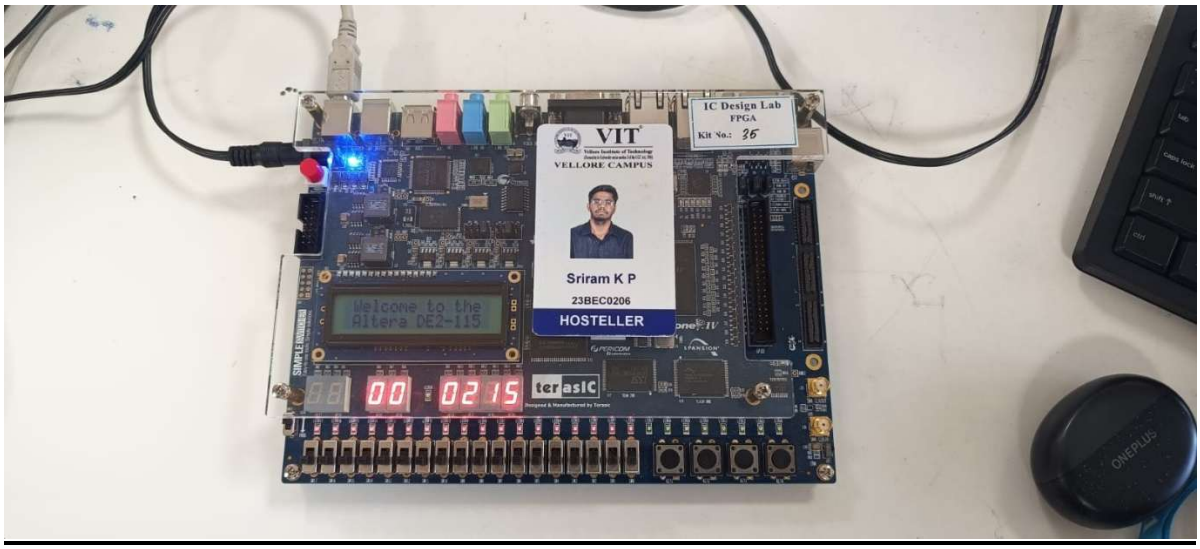


QUARTUS PRIME COMPILATION REPORT :

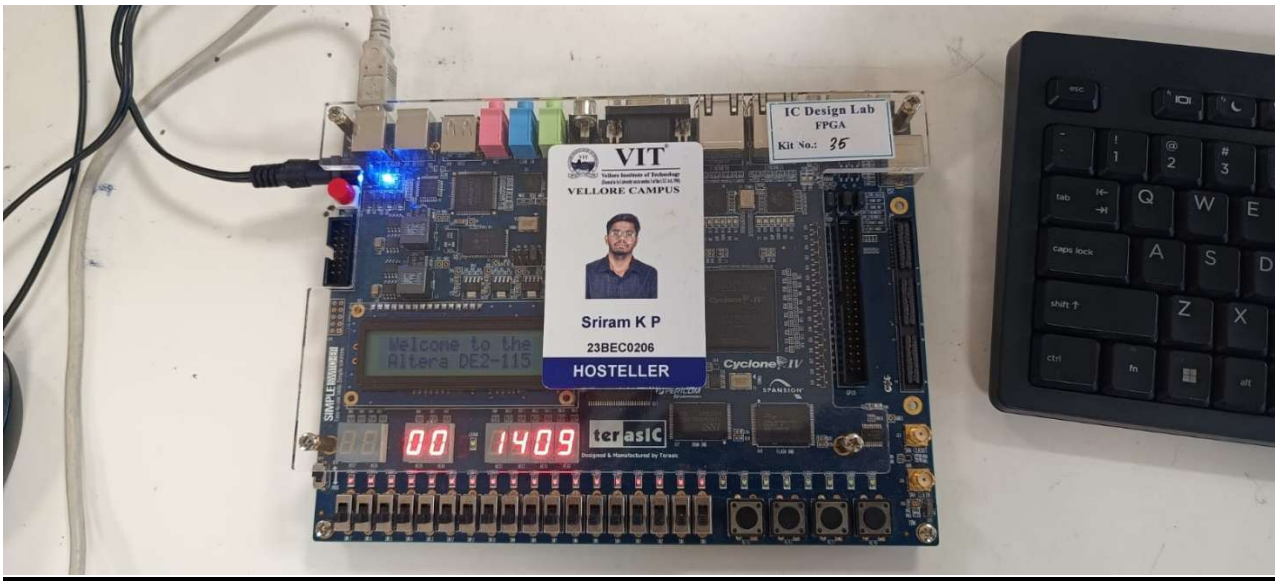


FPGA IMPLEMENTED PHOTOS :

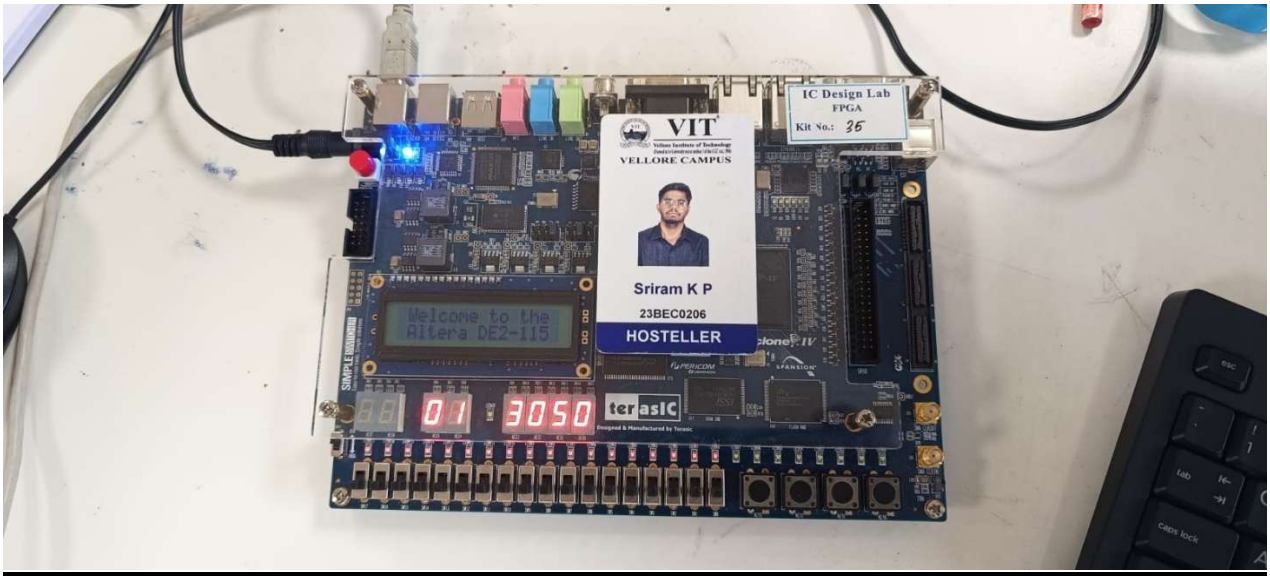
I)



II)


IC Design Lab
FPGA
Kit No.: 35
VIT
VELLORE CAMPUS
Sriram K P
23BEC0206
HOSTELLER
Welcome to the
Altera DE2-115
00 1409
terasIC

III)


IC Design Lab
FPGA
Kit No.: 35
VIT
VELLORE CAMPUS
Sriram K P
23BEC0206
HOSTELLER
Welcome to the
Altera DE2-115
01 3050
terasIC

PART 2)

ii) Implement Synchronous FIFO using Quartus and verify the functionality. Verify the timing analysis and power analysis. Check for the set and hold violation and find the critical path. Attach the RTL view of FIFO and Screenshots of code and outputs.

Verilog RTL Design of Synchronous FIFO :

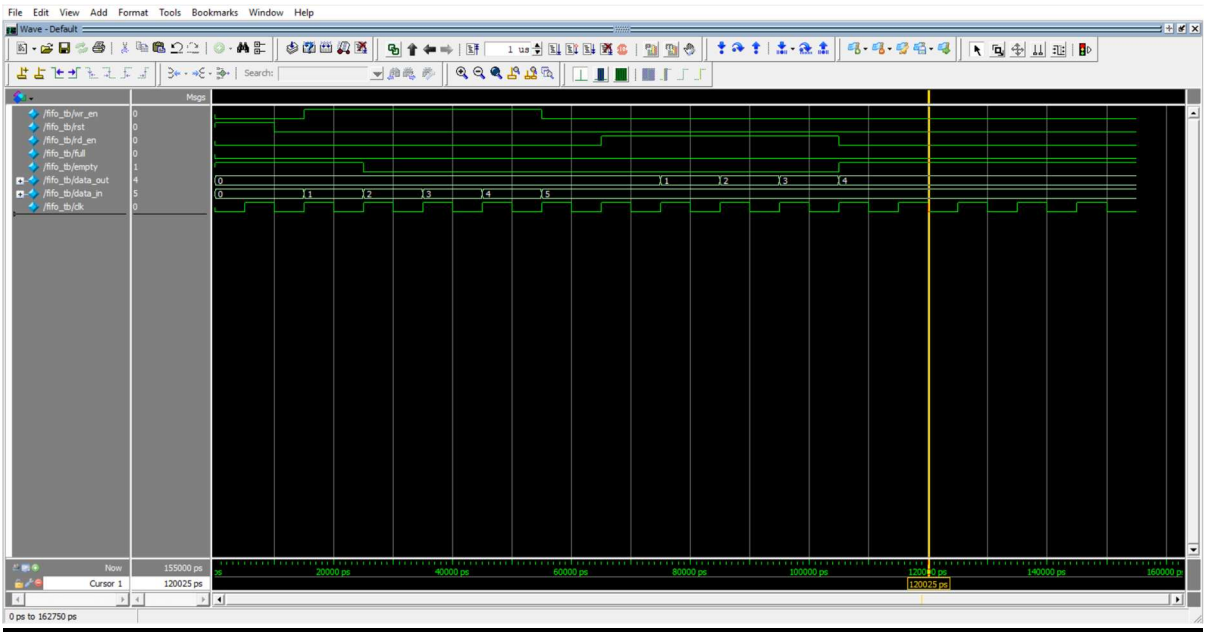
```
module fifo #(
   parameter WIDTH = 8,
   parameter DEPTH = 16)(
   input clk,
   input rst,
   input wr_en,
   input rd_en,
   input [WIDTH-1:0] data_in,
   output reg [WIDTH-1:0] data_out,
   output full,
   output empty);
reg [WIDTH-1:0] mem [0:DEPTH-1];
reg [$clog2(DEPTH)-1:0] wr_ptr, rd_ptr;
reg [$clog2(DEPTH):0] count;
always @(posedge clk or posedge rst) begin
   if (rst)
      wr_ptr <= 0;
   else if (wr_en && !full) begin
      mem[wr_ptr] <= data_in;
      wr_ptr <= wr_ptr + 1;
```

```
end
end
always @(posedge clk or posedge rst) begin
    if (rst) begin
        rd_ptr <= 0;
        data_out <= 0;
    end
    else if (rd_en && !empty) begin
        data_out <= mem[rd_ptr];
        rd_ptr <= rd_ptr + 1;
    end
end
always @(posedge clk or posedge rst) begin
    if (rst)
        count <= 0;
    else begin
        case ({wr_en && !full, rd_en && !empty})
            2'b10: count <= count + 1; // write
            2'b01: count <= count - 1; // read
            default: count <= count;
        endcase
    end
end
assign full  = (count == DEPTH);
assign empty = (count == 0);
endmodule
`timescale 1ns/1ps
module fifo_tb;
```

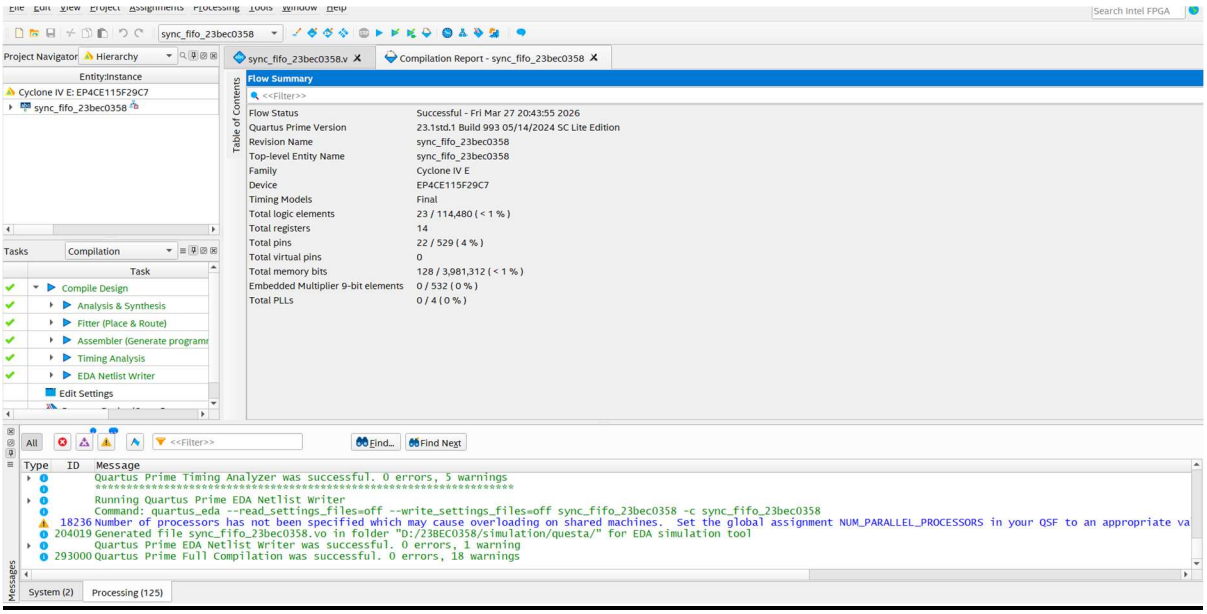
```
reg clk;
reg rst;
reg wr_en, rd_en;
reg [7:0] data_in;
wire [7:0] data_out;
wire full, empty;
fifo uut (
    .clk(clk),
    .rst(rst),
    .wr_en(wr_en),
    .rd_en(rd_en),
    .data_in(data_in),
    .data_out(data_out),
    .full(full),
    .empty(empty));
always #5 clk = ~clk;
initial begin
    clk = 0;
    rst = 1;
    wr_en = 0;
    rd_en = 0;
    data_in = 0;
    #10 rst = 0;
    repeat(5) begin
        @(posedge clk);
        wr_en = 1;
        data_in = data_in + 1;
    end
```

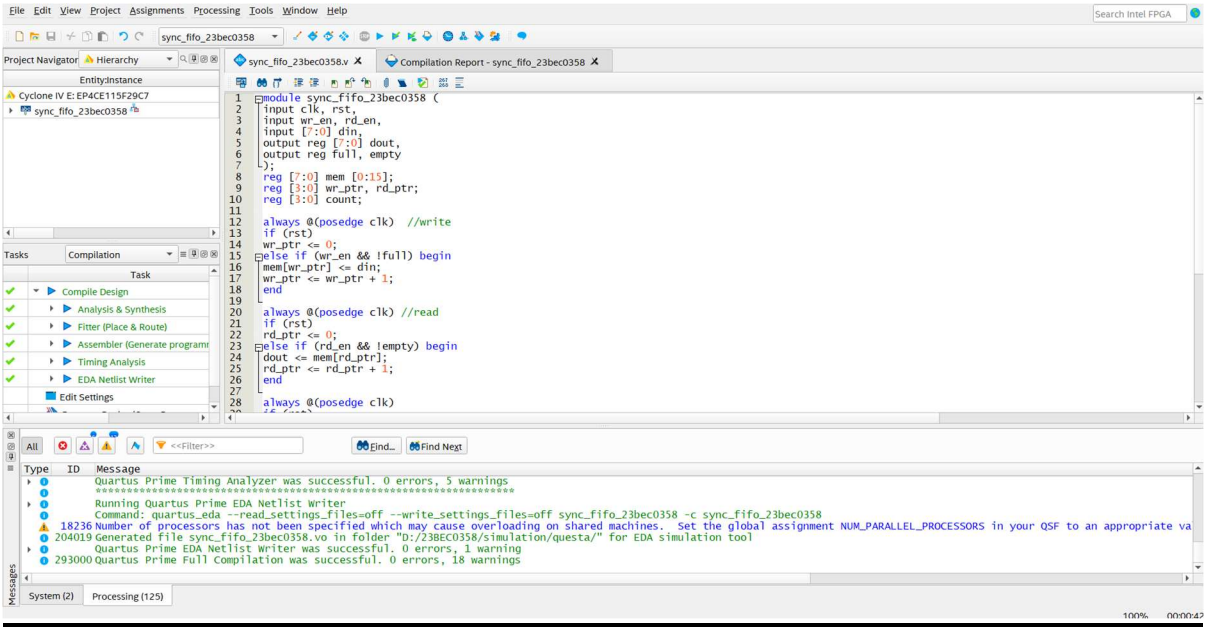
```
    wr_en = 0;
    repeat(5) begin
        @(posedge clk);
        rd_en = 1;
    end
    rd_en = 0;
    #50 $stop;
end
endmodule
```

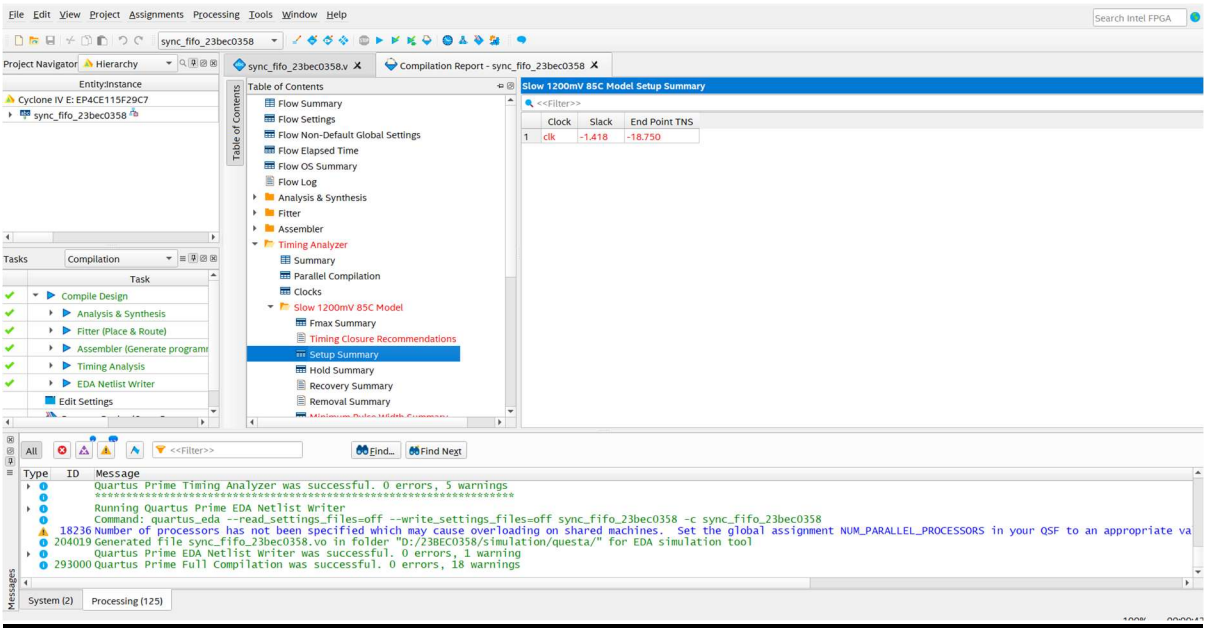
Functional Simulation of Synchronous FIFO using ModelSim :

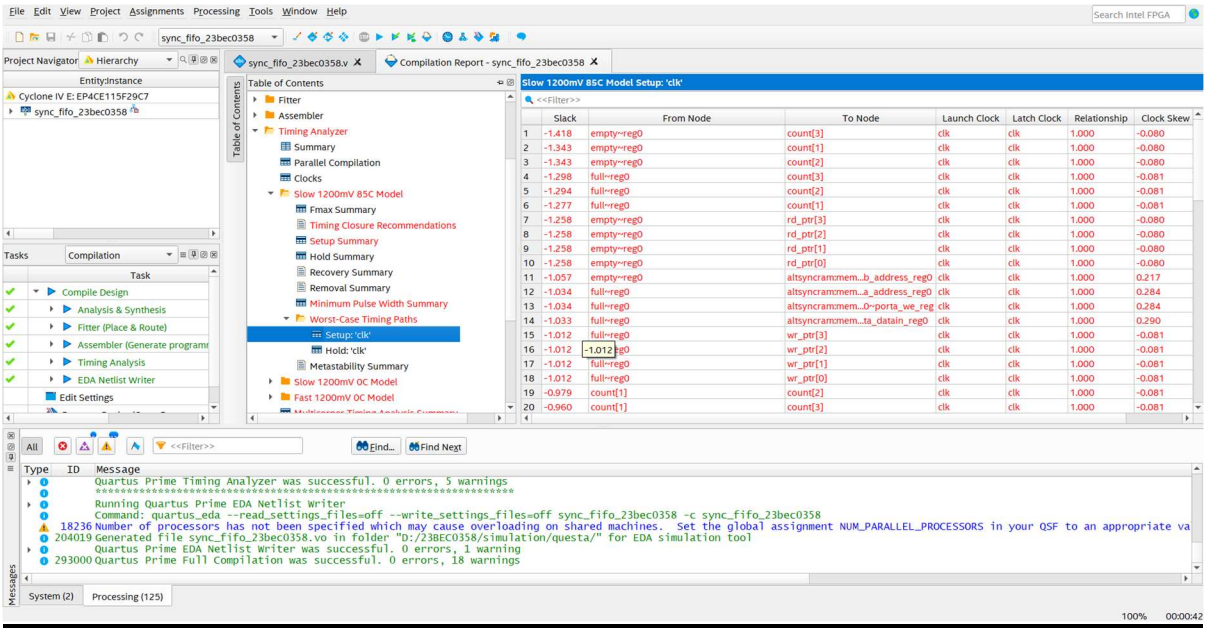


QUARTUS PRIME COMPILATION REPORT :[before Timing Analysis]

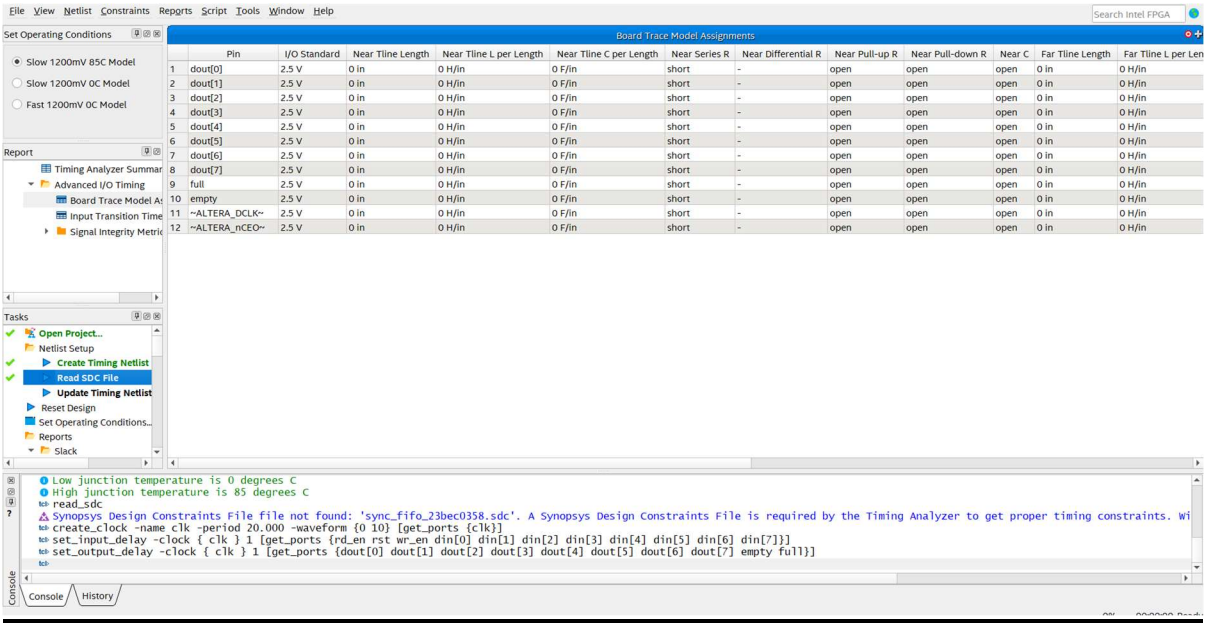


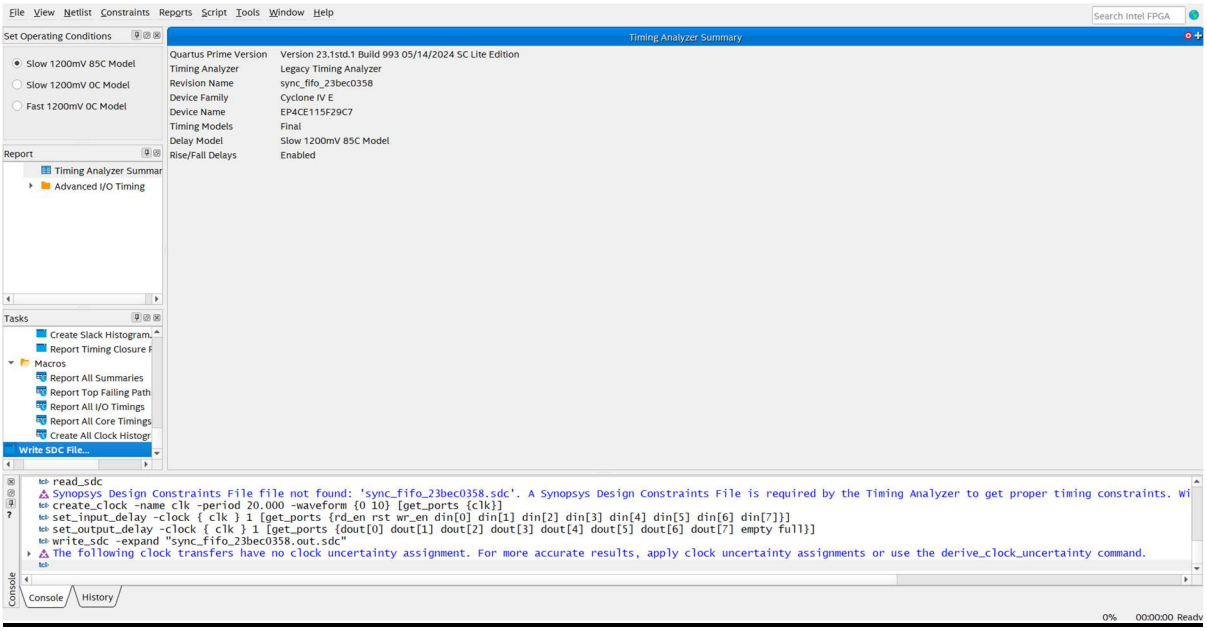



File Edit View Project Assignments Processing Tools Window Help
Search Intel FPGA
sync_fifo_23bec0358
Project Navigator Hierarchy
Entity:Instance
Cyclone IV E: EP4CE115F29C7
sync_fifo_23bec0358
sync_fifo_23bec0358.v
Compilation Report - sync_fifo_23bec0358
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Slow 1200mV 85C Model
Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Recovery Summary
Removal Summary
Slow 1200mV 85C Model Setup Summary
Clock Slack End Point TNS
1 clk -1.418 -18.750
Tasks Compilation
Task
Compile Design
Analysis & Synthesis
Fitter (Place & Route)
Assembler (Generate programm
Timing Analysis
EDA Netlist Writer
Edit Settings
Find... Find Next
Type ID Message
Quartus Prime Timing Analyzer was successful. 0 errors, 5 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate va
204019 Generated file sync_fifo_23bec0358.vo in folder "D:/23BEC0358/simulation/questa/" for EDA simulation tool
Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning
293000 Quartus Prime Full Compilation was successful. 0 errors, 18 warnings
Messages
System (2)
Processing (125)

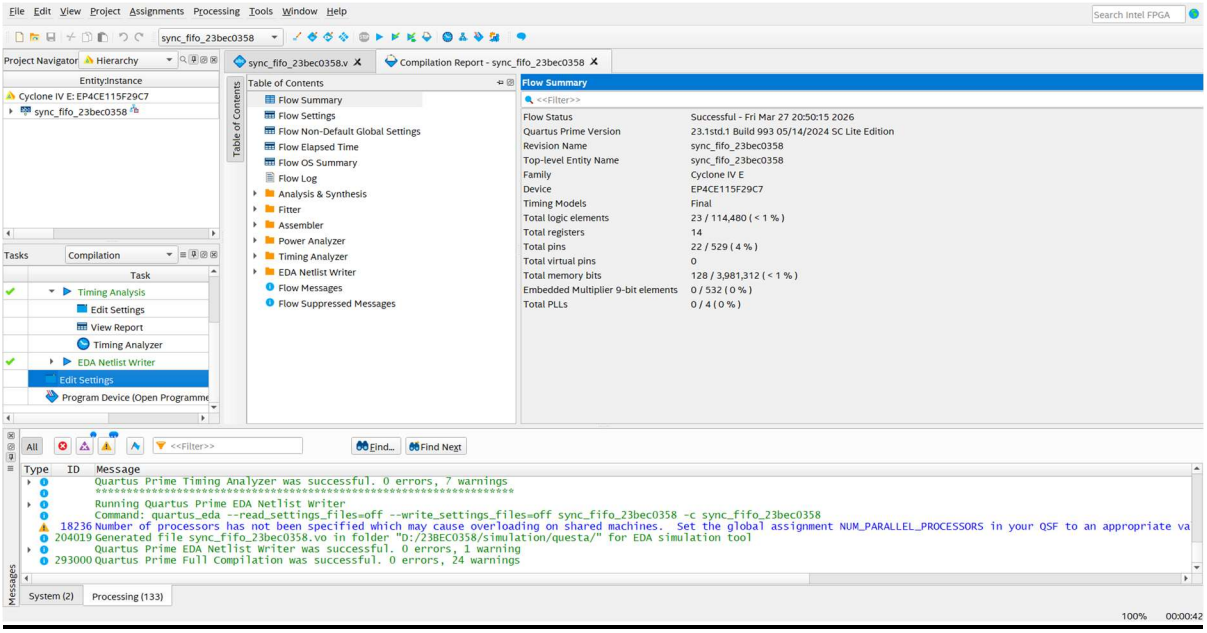

File Edit View Project Assignments Processing Tools Window Help
Search Intel FPGA
sync_fifo_23bec0358
Project Navigator Hierarchy
Entity:Instance
Cyclone IV E: EP4CE115F29C7
sync_fifo_23bec0358
sync_fifo_23bec0358.v
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Clocks
Slow 1200mV 85C Model
Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Recovery Summary
Removal Summary
Minimum Pulse Width Summary
Worst-Case Timing Paths
Setup: 'clk'
Hold: 'clk'
Metastability Summary
Slow 1200mV 0C Model
Fast 1200mV 0C Model
Slow 1200mV 85C Model Setup: 'clk'
Slack From Node To Node Launch Clock Latch Clock Relationship Clock Skew
1 -1.418 empty~reg0 count[3] clk clk 1.000 -0.080
2 -1.343 empty~reg0 count[1] clk clk 1.000 -0.080
3 -1.343 empty~reg0 count[2] clk clk 1.000 -0.080
4 -1.298 full~reg0 count[3] clk clk 1.000 -0.081
5 -1.294 full~reg0 count[2] clk clk 1.000 -0.081
6 -1.277 full~reg0 count[1] clk clk 1.000 -0.081
7 -1.258 empty~reg0 rd_ptr[3] clk clk 1.000 -0.080
8 -1.258 empty~reg0 rd_ptr[2] clk clk 1.000 -0.080
9 -1.258 empty~reg0 rd_ptr[1] clk clk 1.000 -0.080
10 -1.258 empty~reg0 rd_ptr[0] clk clk 1.000 -0.080
11 -1.057 empty~reg0 altsyncram:mem..b_address_reg0 clk clk 1.000 0.217
12 -1.034 full~reg0 altsyncram:mem..a_address_reg0 clk clk 1.000 0.284
13 -1.034 full~reg0 altsyncram:mem..0~porta_we_reg clk clk 1.000 0.284
14 -1.033 full~reg0 altsyncram:mem..ta_datain_reg0 clk clk 1.000 0.290
15 -1.012 full~reg0 wr_ptr[3] clk clk 1.000 -0.081
16 -1.012 -1.012 wr_ptr[2] clk clk 1.000 -0.081
17 -1.012 full~reg0 wr_ptr[1] clk clk 1.000 -0.081
18 -1.012 full~reg0 wr_ptr[0] clk clk 1.000 -0.081
19 -0.979 count[1] count[2] clk clk 1.000 -0.081
20 -0.960 count[1] count[3] clk clk 1.000 -0.081
Tasks Compilation
Task
Compile Design
Analysis & Synthesis
Fitter (Place & Route)
Assembler (Generate programm
Timing Analysis
EDA Netlist Writer
Edit Settings
Find... Find Next
Type ID Message
Quartus Prime Timing Analyzer was successful. 0 errors, 5 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate va
204019 Generated file sync_fifo_23bec0358.vo in folder "D:/23BEC0358/simulation/questa/" for EDA simulation tool
Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning
293000 Quartus Prime Full Compilation was successful. 0 errors, 18 warnings
Messages
System (2)
Processing (125)
100% 00:00:42

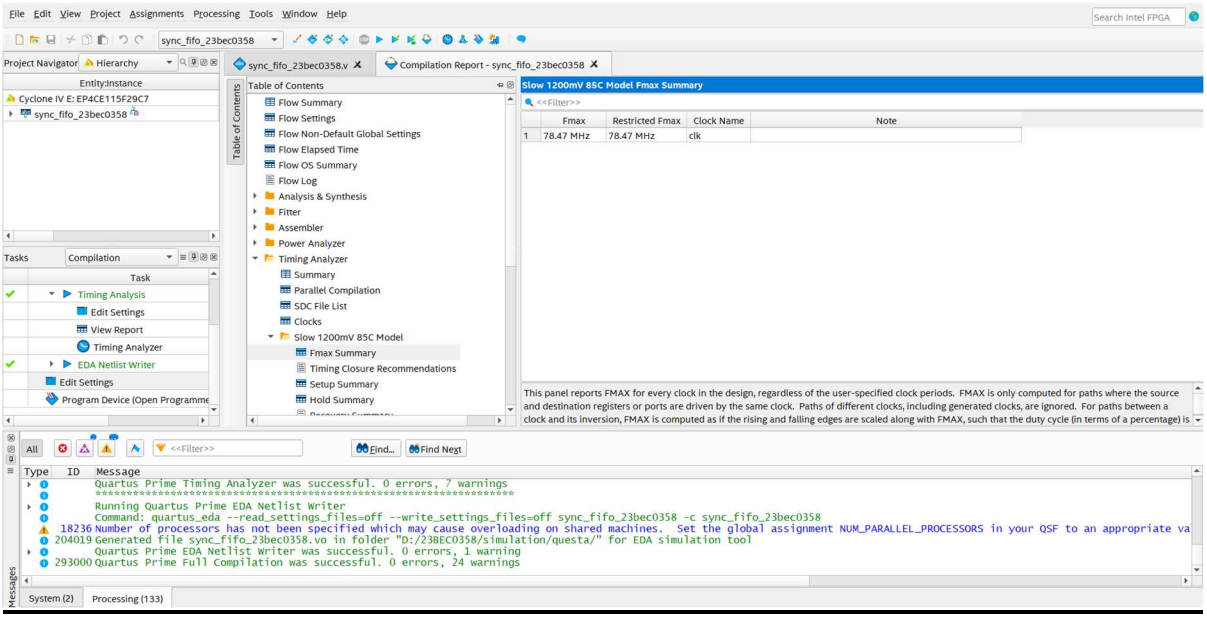
Static Timing Analysis of FIFO using Timing Analyzer :

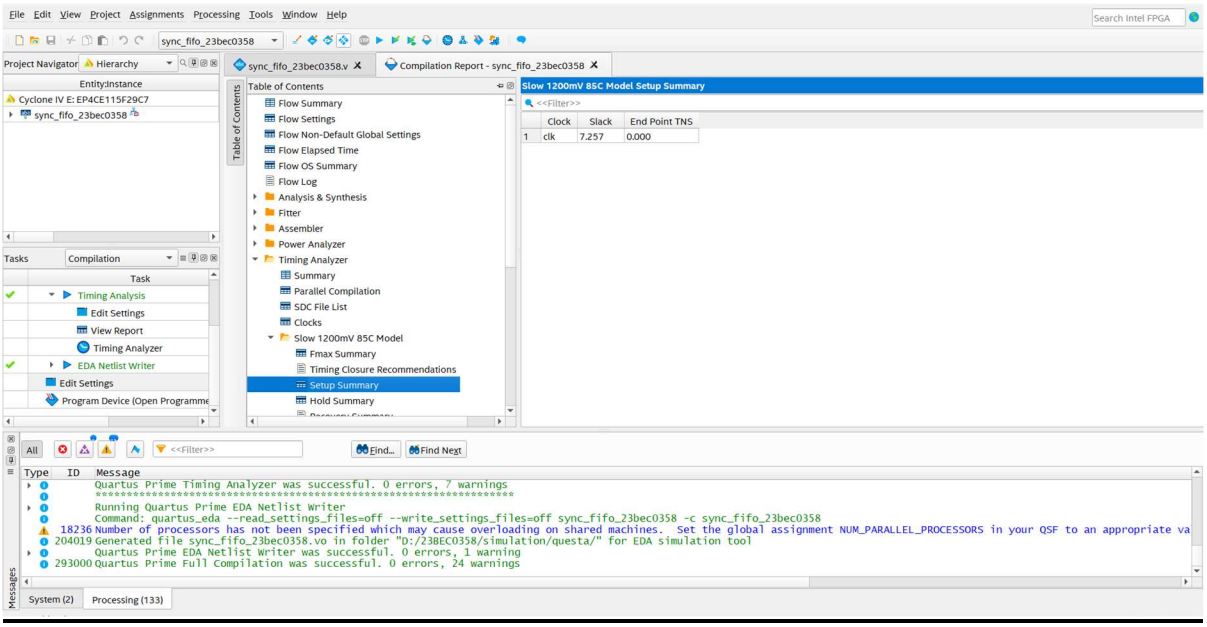


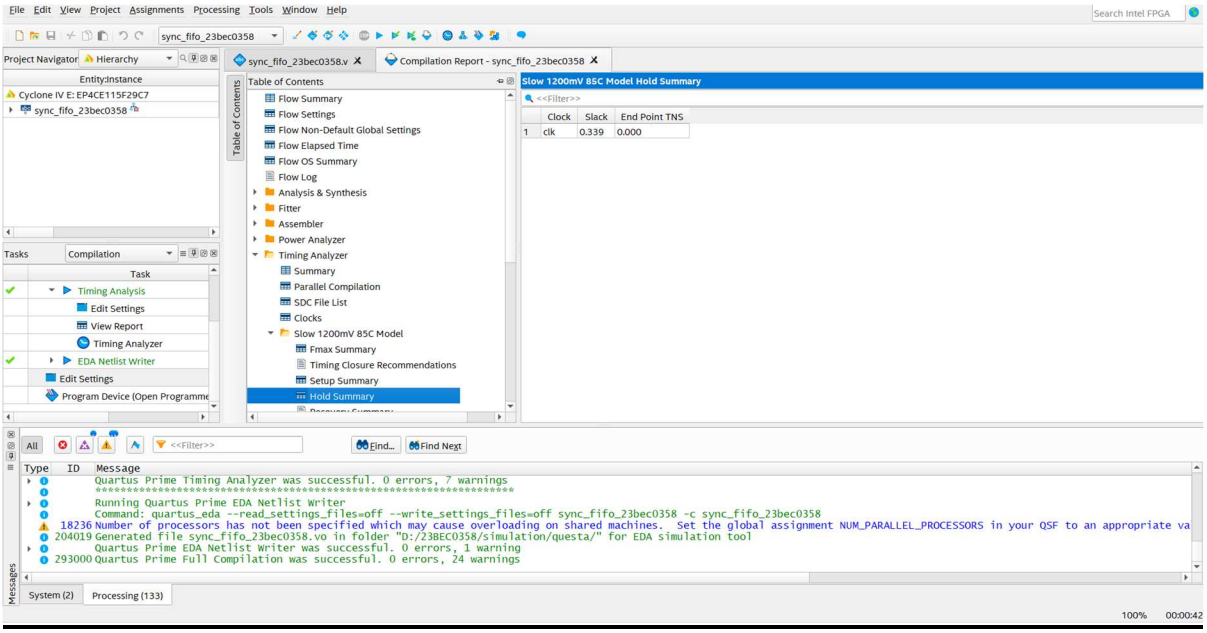


QUARTUS PRIME COMPILATION REPORT :[after Timing Analysis]

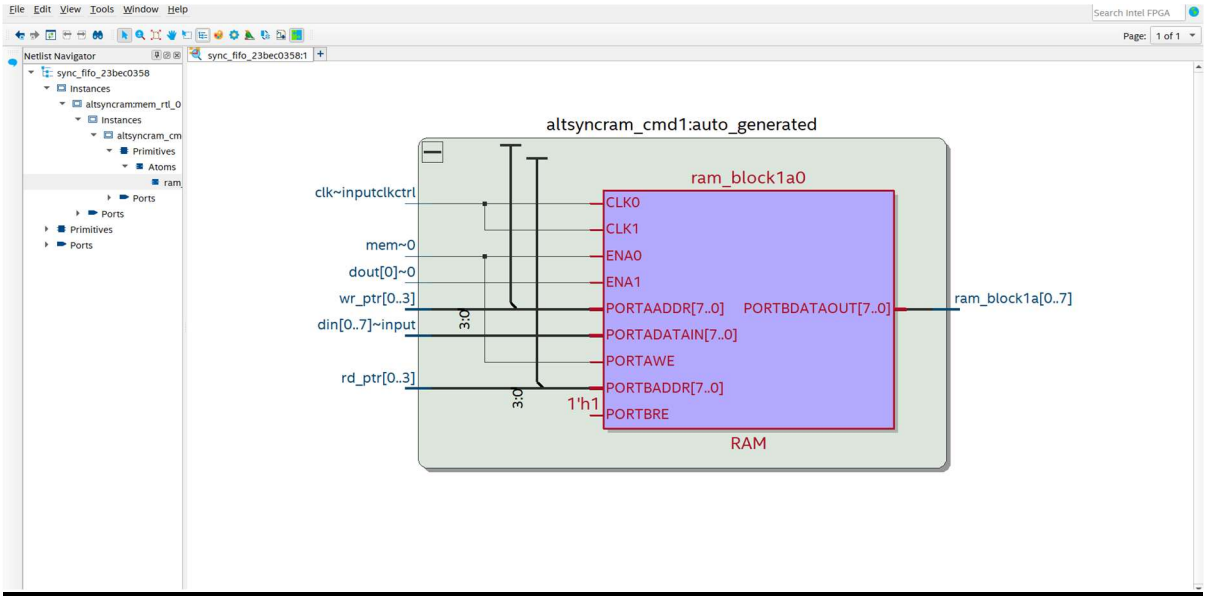




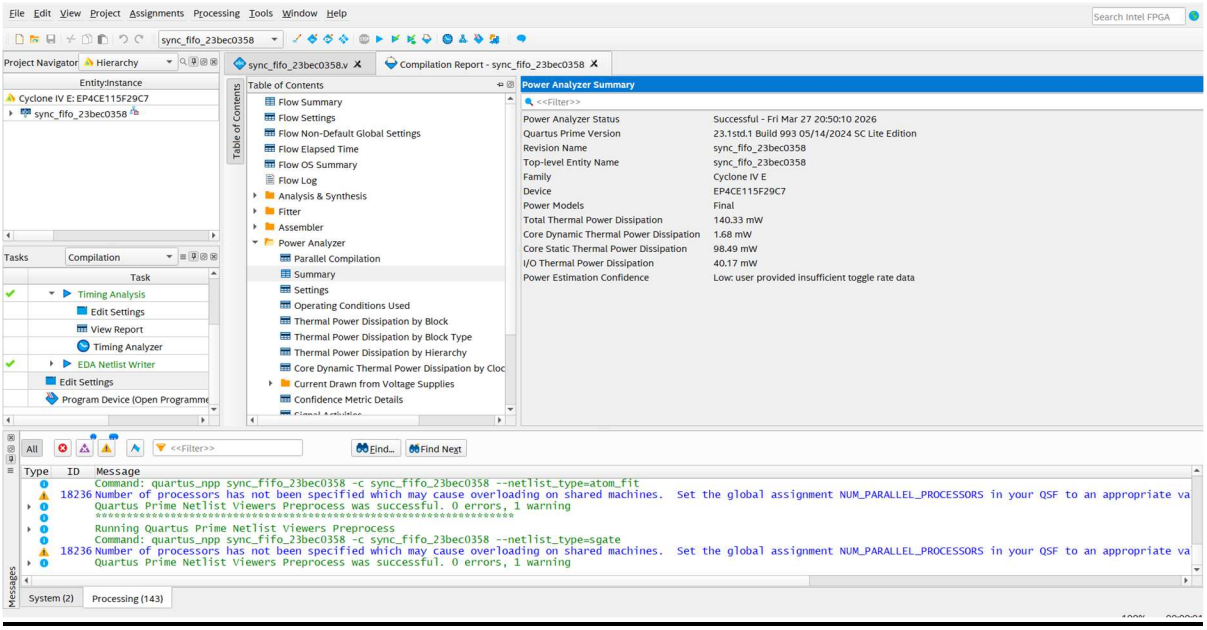

File Edit View Project Assignments Processing Tools Window Help
Search Intel FPGA
sync_fifo_23bec0358
Project Navigator Hierarchy
Entity:Instance
Cyclone IV E: EP4CE115F29C7
sync_fifo_23bec0358
Tasks Compilation
Task
Timing Analysis
Edit Settings
View Report
Timing Analyzer
EDA Netlist Writer
Edit Settings
Program Device (Open Programme
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Slow 1200mV 85C Model
Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Slow 1200mV 85C Model Setup Summary
<<Filter>>
Clock Slack End Point TNS
1 clk 7.257 0.000
All
<<Filter>>
Find... Find Next
Type ID Message
Quartus Prime Timing Analyzer was successful. 0 errors, 7 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate va
204019 Generated file sync_fifo_23bec0358.vo in folder "D:/23BEC0358/simulation/questa/" for EDA simulation tool
Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning
293000 Quartus Prime Full Compilation was successful. 0 errors, 24 warnings
Messages
System (2)
Processing (133)


File Edit View Project Assignments Processing Tools Window Help
Search Intel FPGA
sync_fifo_23bec0358
Project Navigator Hierarchy
Entity:Instance
Cyclone IV E: EP4CE115F29C7
sync_fifo_23bec0358
Tasks Compilation
Task
Timing Analysis
Edit Settings
View Report
Timing Analyzer
EDA Netlist Writer
Edit Settings
Program Device (Open Programme
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Compilation Report - sync_fifo_23bec0358
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SDC File List
Clocks
Slow 1200mV 85C Model
Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Slow 1200mV 85C Model Hold Summary
<<Filter>>
Clock Slack End Point TNS
1 clk 0.339 0.000
All
<<Filter>>
Find... Find Next
Type ID Message
Quartus Prime Timing Analyzer was successful. 0 errors, 7 warnings
Running Quartus Prime EDA Netlist Writer
Command: quartus_eda --read_settings_files=off --write_settings_files=off sync_fifo_23bec0358 -c sync_fifo_23bec0358
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293000 Quartus Prime Full Compilation was successful. 0 errors, 24 warnings
Messages
System (2)
Processing (133)
100%
00:00:42

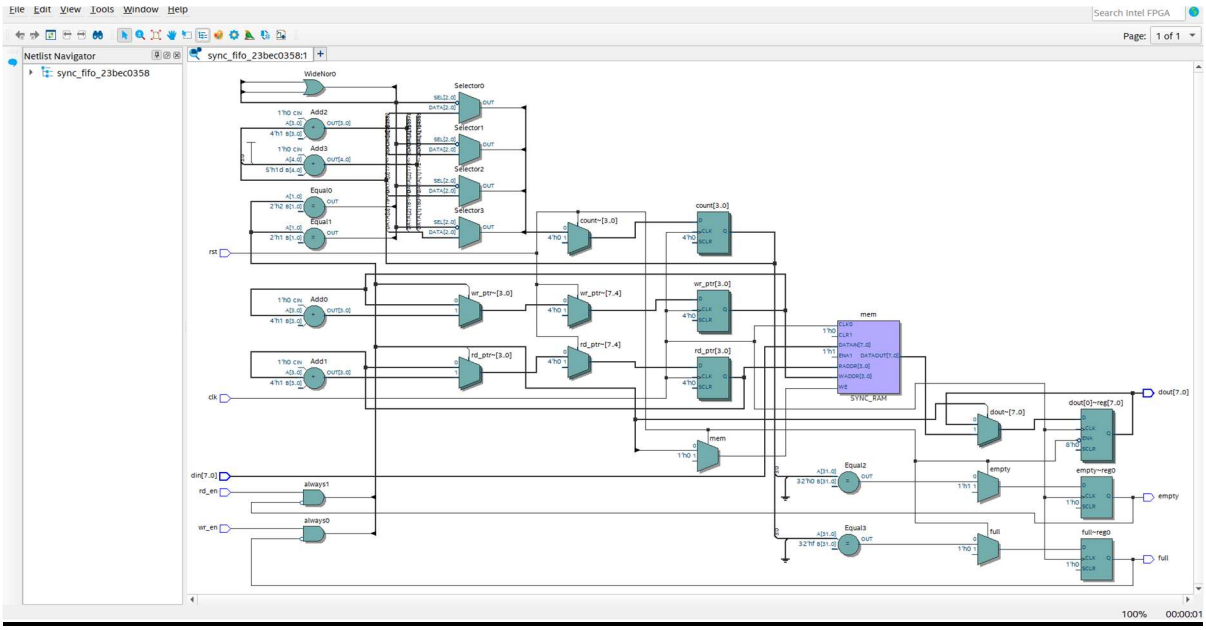
Critical Path Identification in FIFO Design :



Power Analysis of FIFO using Quartus Power Analyzer :



RTL View of FIFO Design :



INFERENCE :

The digital clock was implemented using a clock divider to display time in HH:MM:SS format on a seven segment display. Simulation results confirmed correct time progression, and the design operated reliably using synchronous logic.

The synchronous FIFO was verified through simulation, showing correct FIFO behavior with proper read and write operations. Timing analysis initially showed a violation, but after optimization, positive slack values confirmed reliable operation. The design achieved an operating frequency of about 78.47 MHz and consumed approximately 140 mW of power, indicating efficient performance.